Qn. Set Code-7 (SOT)

Semester: 6th
Programme: B.Tech
Branch: CSE, IT, CSCE, CSSE

SPRING END SEMESTER EXAMINATION-2024

6th Semester B.Tech

CLOUD COMPUTING

IT 3022

(For 2021 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions: [1 × 10]

(a) How does scalability enhances the features of Cloud computing infrastructure?

(b) Distributed computing is a better option for banking services. Justify the statement.

(c) State a real life example where an Infrastructure-as-a-Service is used.

(d) Define Xen Hypervisor.

(e) State at least two advantages of multi-tenancy in cloud computing.

(f) Discuss the application of cloud computing in health sector.

(g) State the category of IP hash method in load balancing.

(h) How is Microsoft Azure different from other cloud provider?

(i) Write the advantages and disadvantages of FCFS scheduling.

(j) List at least one commercially available community cloud.

SECTION-B

2. (a) State and contrast the characteristics of different service models of cloud citing example for each. [5]

   (b) Cluster computing in cloud computing involves multiple nodes or computers to form a single unit. Justify. [5]

3. (a) Explain in details the different categories of virtualization. How does Block virtualization differ from file virtualization? [5]

   (b) Elaborate the storage solutions available in Amazon Web Services. [5]

4. (a) Data centers are used for storage purposes in Cloud computing infrastructure. State the benefits of having a data center. Also explain different privacy issues associated with the same when data is dispersed geographically. [5]

   (b) Trust and security have prevented businesses from fully accepting cloud platforms. To protect clouds, providers must first secure virtualized datacenter resources, uphold user privacy, and preserve data integrity. Justify the statement. [5]

5. (a) What is Live Migration in cloud computing? State the process of Virtual Machine provisioning in virtual machine life cycle. [5]

(b) A company currently works on centralized server mechanism. But with increasing workload it wants to distribute the workload among other servers. Explain the type of scaling require for the same. Mention its characteristics and benefits over other. [5]

6. Write short notes on the following (Any two).

(a) Application of cloud computing in business [5]

(b) Dynamic load balancing in cloud computing [5]

(c) Integration of cloud with Wireless Sensor Network [5]

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